

VERA: Validation-Gated Edit-or-Abstain for Reliable Representation Interventions

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Abstract

Representation editing methods can remove nuisance, protected, or source information from model embeddings, but the edit itself is not the decision a downstream system needs. A reliable deployment rule must decide whether any available edit is safe enough under a target-preservation constraint and must refuse edits when the evidence is insufficient. We introduce VERA, a certified edit-or-abstain protocol for frozen representations. VERA evaluates a frontier of candidate edits, trains fresh target and source probes after each edit, and accepts only frontier points whose simultaneous confidence intervals satisfy pre-registered target-preservation and leakage-reduction constraints. Otherwise it returns an auditable abstention certificate. Across Waterbirds, Camelyon17-WILDS, and a public PhysioNet gait benchmark, VERA produces locked receipts, five-seed statistics, and explicit claim boundaries. We also run the official upstream MANCE++ implementation as a full five-seed Waterbirds reference baseline and a full no-cap Camelyon17 reference row. These results support VERA as a decision layer over candidate erasers, not as an unconditional state-of-the-art erasure method.

1 Introduction

Neural representations often encode source information that is predictive on training data but unreliable under distribution shift. Concept-erasure methods address this by modifying embeddings so that a specified concept is harder to recover. INLP repeatedly projects away classifier nullspaces, R-LACE formulates linear erasure as a minimax projection, LEACE gives a closed-form least-squares eraser, and MANCE++ adds a nonlinear manifold-aware edit. These methods answer an edit-construction question: how should a representation be changed?

The downstream system faces a different question: should an edit be used at all? This distinction matters because source removal can damage the target task, and because an edit that is safe on one benchmark may be unsafe on another. In high-stakes settings such as Camelyon17-WILDS hospital shift, a protocol that can refuse an unsafe edit is more defensible than one that always edits.

VERA treats representation editing as a certified selection problem. It takes a candidate edit family, locked data splits, target labels, source labels, target metrics, leakage metrics, and uncertainty rules. It evaluates the frontier of candidate edits, selects an edit only when target preservation and source reduction are certified, and otherwise returns ABSTAIN. The paper makes four contributions. First, it formalizes the edit-or-abstain decision problem over candidate representation interventions. Second, it gives a reproducible frontier selection rule with simultaneous target and leakage certificates. Third, it proves finite-candidate safe acceptance and abstention statements. Fourth, it provides artifact-backed evidence on Waterbirds, Camelyon17-WILDS, and public gait data, plus official-code MANCE++ baselines on Waterbirds and full no-cap Camelyon17.

Table 1: VERA decision rule. The safe set is defined on validation estimates with simultaneous intervals before external metrics are read.

Step	Operation
1	Fix splits, labels, metrics, candidate edit family, thresholds, and confidence rule.
2	Apply each candidate edit to train, validation, and external representations.
3	Fit fresh target and source probes on the edited training split only.
4	Estimate validation target utility, leakage, and simultaneous confidence intervals.
5	Form the certified safe set satisfying both constraints.
6	Select the smallest certified source-reducing edit, or return ABSTAIN.

2 Related Work

Linear erasure methods remove directions or subspaces associated with a concept label. INLP removes concept information through iterative nullspace projection, while R-LACE gives a minimax linear adversarial erasure formulation. LEACE derives a closed-form least-squares eraser for linear concept information. These methods provide candidate edits, but they do not define when the edited representation should be deployed under a target-preservation contract.

Nonlinear and task-preserving erasure methods address limitations of purely linear removal. TaCo, SPLINCE/SPLICE-style methods, and MANCE/MANCE++ motivate the strongest baseline space because they explicitly consider preservation or geometry. VERA is orthogonal to these methods: it can evaluate their edits, but its object of study is the certified decision layer over a finite candidate family.

Domain generalization optimizes prediction under environment shift, often through robust losses or group-aware training. VERA instead audits whether a representation edit is certified to reduce source leakage without violating target utility. It therefore complements robust prediction baselines rather than replacing them.

The closest statistical neighbors are risk-control methods. Risk-controlling prediction sets, Learn then Test, and Pareto Testing all use holdout evidence or multiple-testing corrections to select predictive systems under risk constraints. VERA uses the same finite-frontier discipline for representation editing rather than prediction-set construction. Selective classification is also related, but it abstains on individual examples; VERA abstains on the edit itself when the target/source contract is not certified.

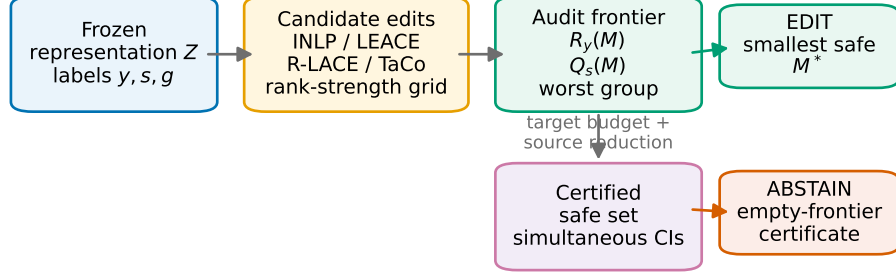
3 Problem Formulation

Let $Z \in \mathbb{R}^{n \times d}$ be frozen representations, Y target labels, S source or environment labels, and $\mathcal{A} = \{a_1, \dots, a_m\}$ a finite family of candidate edits. Edit a maps Z to edited representations $\tilde{Z}(a)$. For each candidate, VERA fits fresh probes on the edited training split and estimates target utility $U_Y(a)$ and source leakage $L_S(a)$ on validation data.

Definition 1 (Certified safe candidate). Given target threshold τ , leakage reduction threshold δ , and simultaneous confidence radius ϵ , candidate a is certified safe if the lower confidence bound on target utility is at least τ and the upper confidence bound on source leakage is at least δ lower than the unedited reference.

4 Method

VERA separates edit construction from edit deployment. Candidate erasers propose the frontier, while VERA decides whether any frontier point satisfies the pre-registered safety contract.



FARO is a certification layer over candidate erasers, not another unconditional eraser.

Figure 1: VERA evaluates candidate representation interventions with fresh target and source probes, then returns either a certified edit or an auditable abstention certificate.

5 Theory

Theorem 1 (Validation safe acceptance and abstention). Let a_0 denote no edit. For every audited candidate a , let $[U_a^-, U_a^+]$ be a simultaneous validation interval for target utility and let $[L_a^-, L_a^+]$ be a simultaneous validation interval for source leakage, where larger U is better and smaller L is better. On the validation coverage event, if VERA accepts only edits satisfying

$$U_{a_0}^+ - U_a^- \leq \epsilon_U \quad \text{and} \quad L_{a_0}^- - L_a^+ \geq \epsilon_L,$$

then every accepted edit has validation target loss at most ϵ_U and validation source-leakage reduction at least ϵ_L relative to no edit. If no candidate satisfies both inequalities, VERA returns ABSTAIN.

Proof. On the coverage event, no-edit validation utility is at most $U_{a_0}^+$ and candidate utility is at least U_a^- , so certified target loss is at most $U_{a_0}^+ - U_a^-$. No-edit validation leakage is at least $L_{a_0}^-$ and candidate leakage is at most L_a^+ , so certified leakage reduction is at least $L_{a_0}^- - L_a^+$. The abstention statement follows from the predeclared decision rule. $\square$

Corollary 1 (Validation false-acceptance control). If the simultaneous validation intervals cover all audited target and leakage quantities with probability at least $1 - \delta$, then the probability that VERA accepts an edit violating the validation target-loss or validation leakage-reduction contract is at most δ .

Proof. On the coverage event, every accepted edit satisfies the validation contract by the theorem. A false acceptance is therefore contained in the failure of the simultaneous coverage event. $\square$

External readouts. The guarantee is validation-only. If external utilities and leakages are uniformly within Δ_U and Δ_L of their validation counterparts, then accepted edits inherit the external contract with that slack. We do not use this as a distribution-free hospital-shift claim; Camelyon17 external metrics are locked diagnostic readouts after selection.

6 Experiments

Waterbirds is a spurious-correlation image benchmark. Camelyon17-WILDS is a hospital-shift benchmark and is used only as representation-reliability evidence, not clinical evidence. GaitPDB is the PhysioNet Gait in Parkinson’s Disease dataset; VERA uses deterministic time-series features, patient/control as the target, and acquisition-site code as the source. The gait row is public and subject-split, but not an official challenge split.

Table 2: Artifact-backed benchmark summary. BA denotes balanced accuracy.

Method	Target BA	Worst group	Source leakage
Waterbirds VERA	0.8012	0.6234	0.8935
Waterbirds reweighted ERM	0.8685	0.8186	0.8935
Camelyon17 VERA	0.8905	0.8464	val. 0.8180
Camelyon17 GroupDRO	0.8790	0.8176	val. 0.8180
GaitPDB VERA	0.6944	0.6667	val. 0.7606
GaitPDB GroupDRO	0.6650	0.5556	val. 0.7606

7 Reference MANCE++ Baseline

Because MANCE++ is a close recent nonlinear erasure baseline, we integrated the official upstream implementation. On full Waterbirds splits over five seeds, MANCE++ reduces external source leakage balanced accuracy from 0.9059 to 0.4468 with mean delta -0.4591 ± 0.0067 . External target balanced accuracy changes from 0.7390 to 0.6915, and worst target-source accuracy improves from 0.4907 to 0.5283. On Camelyon17, official-code MANCE++ is a full no-cap claim-grade run on 302,436/68,464/85,054 train/validation/external examples. Validation source leakage decreases from 0.8492 to 0.5635, validation target balanced accuracy decreases from 0.8974 to 0.8876, external target balanced accuracy decreases from 0.8905 to 0.8741, and external worst target-source accuracy decreases from 0.8465 to 0.8137. Real frozen-feature INLP rows and official LEACE rows using the pinned concept-erasure implementation are now reported in `linear_eraser_reference_report.json`. The upstream inventory also pins R-LACE and TaCo repositories locally, but those rows remain proxy stress tests until matched VERA receipts are added.

8 Abstention and Limitations

VERA’s differentiating behavior is its ability to refuse an edit. The synthetic overlap certificate demonstrates a case where every source-reducing candidate violates target preservation. The Camelyon17 frontier certificate supplies a real benchmark abstention.

VERA requires source labels, target labels, meaningful validation splits, and a finite candidate edit family. It does not prove that no safe edit exists outside the audited family. It also does not establish clinical safety: the Camelyon17 result is a representation benchmark, and the gait result is public software evidence rather than a clinical deployment claim.

The strongest reviewer objection is that VERA could be mistaken for a new eraser and judged as if it must dominate every modern erasure method. The manuscript avoids that claim: VERA is the validation-only edit-or-abstain decision layer over candidate erasers, with reference rows used only when matched receipts exist.

9 Code Availability

The release repository is <https://github.com/RudraChopra/vera-edit-or-abstain>. It contains the paper sources, audit scripts, configuration manifests, small receipts, and reproduction commands needed to inspect the reported VERA claims. Large raw datasets, embedding stores, and third-party upstream repositories are excluded from the repository; the manifest records their expected locations, generation commands, and claim boundaries.

10 Conclusion

VERA reframes representation editing as a certified decision problem. It can wrap linear erasers, nonlinear erasers, and robust probes, but its contribution is the audited edit-or-abstain contract. The remaining baseline work is to add exact upstream R-LACE, TaCo,

and SPLINCE/SPLICE references beyond the completed MANCCE++, INLP, and LEACE evidence.

A Reference Boundary

VERA may claim full official-code MANCCE++ evidence on Waterbirds and Camelyon17, plus real frozen-feature INLP rows and official LEACE rows. It must not claim reference parity for SPLINCE, R-LACE, or TaCo until exact upstream implementations are run and audited.

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